

ECE 2201 Microelectronic Circuits I

Lab #2: Diode Applications

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Abstract

This experiment explores the behavior of diode-based rectifier circuits and reveals how the addition of capacitive filtering can improve the quality of DC output. We started with half-wave and full-wave rectifiers using 1N4004 diodes. Through these circuits, we explored how diode behavior influences waveforms and gained an understanding of how capacitive filtering can reduce ripple. Additionally, this lab moves towards regulated supplies by utilizing a Zener diode voltage regulator and an LM7805 linear regulator, which both provide voltage stability under various loads. This experiment examines the measured waveforms, the transformer operation, ripple behavior, and the dropout limits of the voltage regulators.

EQUIPMENT

- ECE Lab Kit
- DC Power Supply
- 12VAC Transformer
- DVM
- Function Generator
- Oscilloscope

Introduction

The purpose of Lab 2 – Diode Applications is to gain an understanding of how AC signals can be transformed into unregulated DC voltages through rectification, filtering, and regulation. This lab starts with the basic half-wave and full-wave rectifier circuits, where the diode behavior can determine the parts of the AC waveform that reach the load. The effects of adding the filter capacitors are then studied and analyzed, showing how capacitor size can directly affect the ripple amplitude and smoothness of the DC output. This lab then goes further and investigates voltage regulation through using a Zener diode-based regulator and an LM7805 linear regulator. The circuits reveal how a regulated power supply can maintain a near constant output voltage as the load resistance is changing. Ultimately, this lab reveals how the differences between unregulated and regulated supplies, as well as demonstrating how each stage can enhance the stability and quality of DC output.

W1

L1 Circuit Description

The rectifier circuit for W1 is built with the 12VAC transformer, a 1N4004 diode that is in series and a $1\text{k}\Omega$ load resistor connected to ground. This diode is oriented to be forward biased during the positive half-cycle which allows current to flow only when the transformer output is positive. The oscilloscope was able to measure VAC1 directly through the transformer and then V_L across the load resistor. This configuration of the circuit produces single-polarity output and blocks all the negative half-cycles.

W1-1.

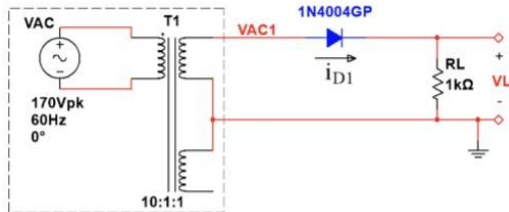


Figure 1.1 Half-Wave Rectifier Circuit (no filter cap)

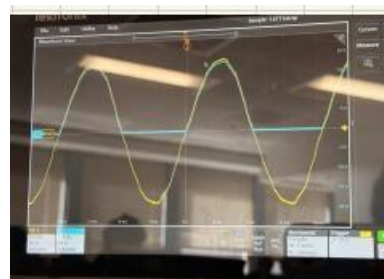


Figure 1.2 Waveform of Figure 1.1

At the positive peak, V_L was approximately 0.7 V lower than VAC1 due to the diode voltage drop. During the negative half-cycle of VAC1, V_L remained near 0 V.

W2

L2 Circuit Description

The rectifier from W1 was modified to have a $10\text{ }\mu\text{F}$ capacitor that is in parallel with the $1\text{k}\Omega$ load resistor. The capacitor has its positive end connected to the node of the diode's output and its negative end connected to the ground. This allows for the capacitor to charge to the peak of the rectified waveform during conduction and then slowly discharge through the load during the non-conduction periods. Channel 1 of the oscilloscope measured the transformer output (VAC1) and channel 2 measured the output voltage across the capacitor and load resistor (R_L).

W2-1.

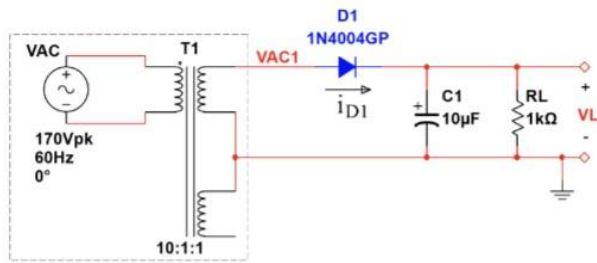


Figure 2.1 Half-Wave Rectifier Circuit (with filter cap)

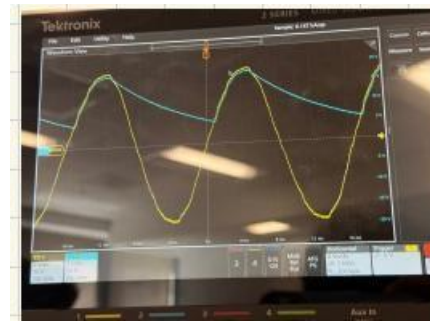


Figure 2.2 Waveform of Figure 2.1 with 10 µF

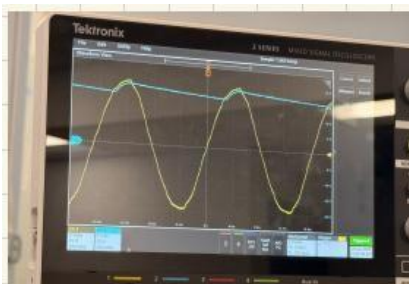


Figure 2.3 Waveform of Figure 2.1 with 100 µF

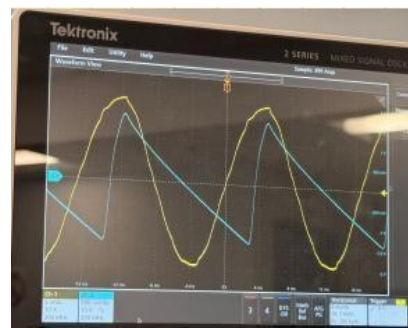


Figure 2.4 AC coupling Waveform of Figure 2.1 with 10 µF

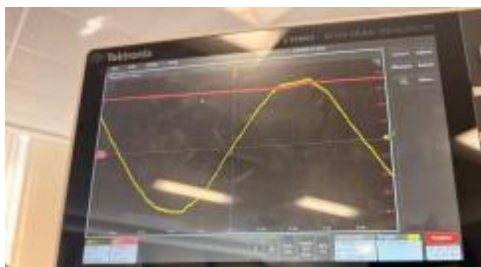


Figure 2.5 Waveform of Figure 2.1 with 330µF

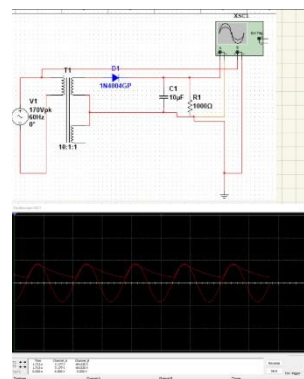


Figure 2.6 Waveform simulation of Figure 2.1 with 10 µF

W2-2.

The ripple voltage in a half-wave rectifier with a filter capacitor is determined through how quickly the capacitor can discharge the load between the conduction peaks. The theoretical relationship is determined through the formula: $V_{RPP} = V_P * \frac{1}{FRC}$. This shows that ripple decreases when capacitance increases and will increase when the load resistance is smaller. With the fixed load resistance of $1k\ \Omega$, measurements were made for several capacitance values. In Figure 2.2 there is a $10\ \mu\text{F}$ capacitor that charges rapidly with a steep upward slope as well as a large ripple amplitude. Then, in Figure 2.3, a $100\ \mu\text{F}$ capacitor was used, which decreased the ripple significantly while the voltage stayed near the peak for a larger amount of the cycle. Also, as shown in Figure 2.6, with an image of the simulated circuit as well as the result, its waveform has a nearly identical shape to the experimental result in Figure 2.2.

W2-3.

To attempt to reduce the ripple to less than $1V_{pp}$, the ripple formula stated above in W2-2 was used. After plugging in the known values into the equation, $1V = 20V * \frac{1}{60\text{Hz} * 1000\ \Omega * C}$, the calculation predicts that a capacitance of $333.333\ \mu\text{F}$ is needed to bring the ripple under $1V_{pp}$. Since the ECE lab kit only provides standard capacitor values, $330\ \mu\text{F}$ was used. After testing in the circuit, the measured waveform confirmed the prediction with the ripple voltage dropping to about $1V_{pp}$ and the output voltage also staying very close to the peak between the conduction intervals. This matches the theory that the larger capacitance will increase the time constant (RC) resulting in the capacitor discharging more slowly through the $1k\ \Omega$ load and dramatically decreasing the ripple.

W3

L3 Circuit Description

The circuit used in W3 is a full-wave rectifier circuit that is built with a center-tapped transformer with two diodes. Each half of the transformer's secondary has alternating polarity causing one diode to conduct during the positive half-cycle and the other diode to conduct during the negative half-cycle in the negative half-cycle. This results in two positive pulses per one AC cycle producing a full-wave rectified output at 120 Hz. With no filter capacitor being used the output voltage will drop to 0V between peaks.

W3-1.

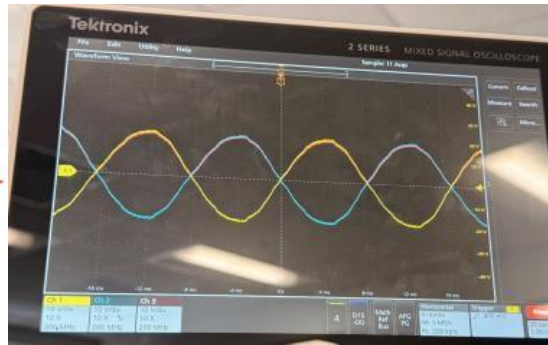
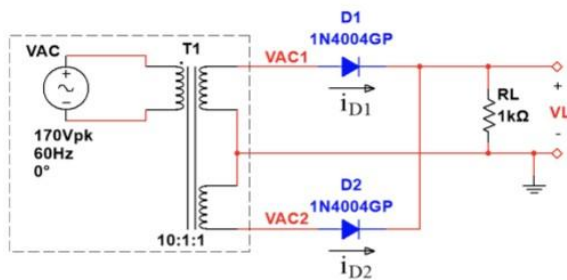


Figure 3.1: Full-Wave Rectifier Circuit (no filter cap) Figure 3.2: Waveform of Figure 3.1 with channel 1 being VAC1 and Channel two being VAC2

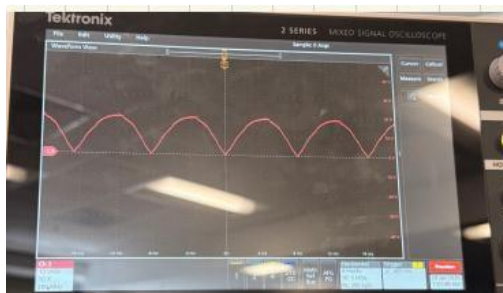


Figure 3.3: Waveform of Figure 3.1 with channel 3 being V_L

Based on the figures above, VAC1 and VAC2 are both sine waves with same amplitude and frequency but are 180 degrees out of phase with one waveform being positive while the other is negative. The full-wave rectifier output produces two positive pulses per cycle which doubles the ripple frequency when compared to the half-wave rectifier. Different than the half-wave case, the output contains a positive pulse on both half-cycles instead of skipping every other half-cycle but still will drop to 0V between peaks because there is no capacitor in this configuration.

W4.

L4 Circuit Description

The full-wave rectifier circuit from W3 was updated with the addition of a filter capacitor in parallel with the 1kΩ load resistor. The positive end of the capacitor was connected to the rectifier's output node, and the negative end was to the ground. This configuration allows the capacitor to charge to the peak of the full-wave rectified signal and then slowly discharge through the load between pulses. Channel 1 of the oscilloscope measured VAC1, channel 2 measured VAC2, and channel 3 measured the output across the capacitor and load resistor.

W4-1.

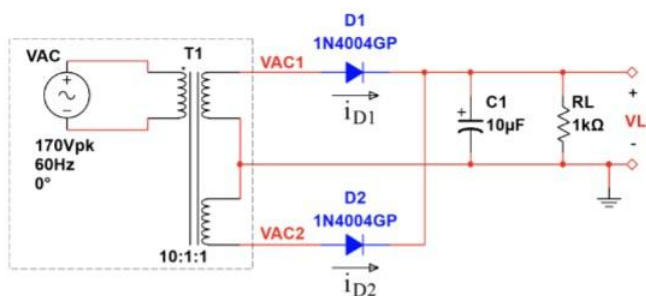


Figure 4.1: Full-Wave Rectifier Circuit (with filter cap)

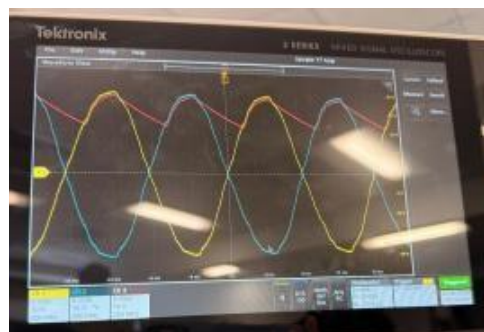


Figure 4.2: Waveform of Figure 4.1 with 10 μF

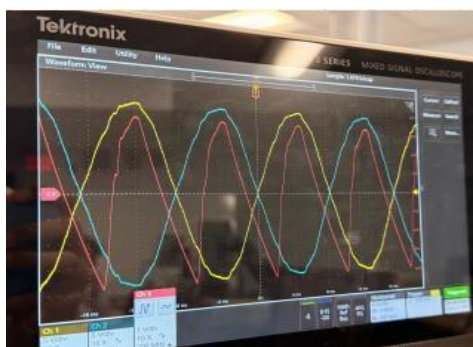


Figure 4.3: AC coupling Waveform of Figure 4.1 with 10 μF

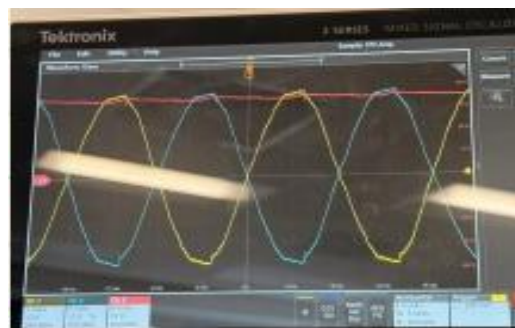


Figure 4.4: Waveform of Figure 4.1 with 200 μF

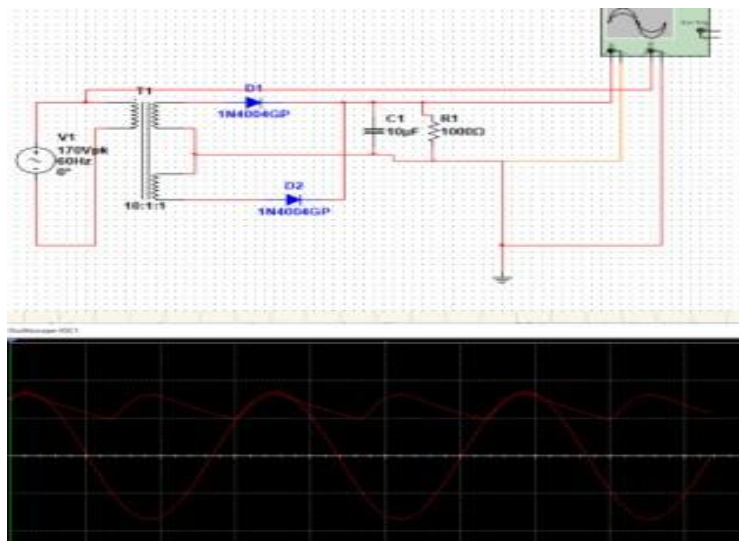


Figure 4.5: Waveform simulation of Figure 4.1 with 10 μF

W4-2.

For a full-wave rectifier with a filter capacitor, the ripple voltage is given by the equation:

$$V_{RPP} = V_P * \frac{1}{2FRC}.$$

This shows that ripple has an inverse relationship with both capacitance and load resistance. Larger capacitors should be able to discharge slower and reduce the ripple and larger load resistors also reduce ripple because less current is drawn from the capacitor. Also, as shown in Figure 4.5, with an image of the simulated circuit as well as the result, its waveform has a nearly identical shape as the experimental result in Figure 4.2.

W4-3.

In the design challenge, a ripple peak to peak of 1V or less is needed. Using the equation from W4-2 and plugging in the known values, $1V = 20V * \frac{1}{2 * 60Hz * 1000 \Omega * C}$, it was calculated that the capacitance needs to be 166.67 μF to achieve a $1V_{pp}$ ripple. Due to the limitations of the ECE lab kit, two 100 μF capacitors in parallel were used yielding 200 μF which is approximately the needed 166.67 μF . The oscilloscope measurements revealed that the ripple was lower than in the 10 μF case and had approximately a $1V_{pp}$. Overall, the measured experimental results as well as the simulated results both support the theoretical prediction that a capacitance of 166.67 μF or higher yields ripple below $1V_{pp}$.

W4-4.

The full-wave rectifier produces two charging pulses per cycle which greatly reduces the ripple with the same load and capacitance applied in comparison to the half-wave circuit. This is because the capacitor is refreshed twice as often and the full-wave design results in a higher, more stable DC output. The main advantage is increased efficiency and smoother voltage whereas the half-wave rectifier is simpler but results in larger ripple and a worse utilization of the AC waveform. Ultimately, the full-wave circuit is better for power supply application with its main drawback being increased circuit complexity.

W5.

L5 Circuit Description

The circuit in L5 is a full-wave rectifier built with a center-tapped transformer and two diodes. Each half of the transformer's secondary alternates the polarity. This allows one diode to conduct in the positive half-cycle and the other to conduct during the negative half-cycle, allowing two positive pulses per cycle with a 120 Hz full-wave rectified signal. It is updated from the circuit in L4, as we added two $510\ \Omega$ resistors in parallel to create an approximately $250\ \Omega$ resistor in series. Next, we swapped out the $10\ \mu\text{F}$ filter capacitor for a $100\ \mu\text{F}$ filter capacitor. We then updated the circuit by adding a Zener diode parallel with the filter capacitor and $1\text{k}\ \Omega$ resistor. The Zener diode voltage regulator will be used to reduce ripple further and produce a better approximation of an ideal constant DC voltage.

W5.1

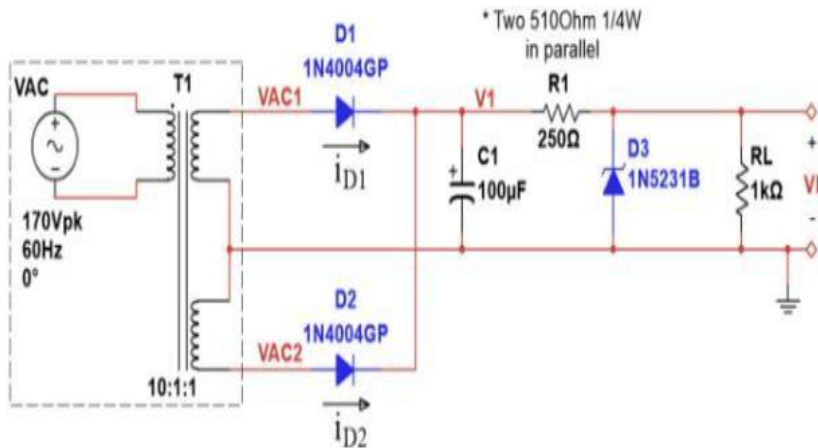


Figure 5.1: Full-Wave Rectifier Circuit (with cap and Zener regulator)

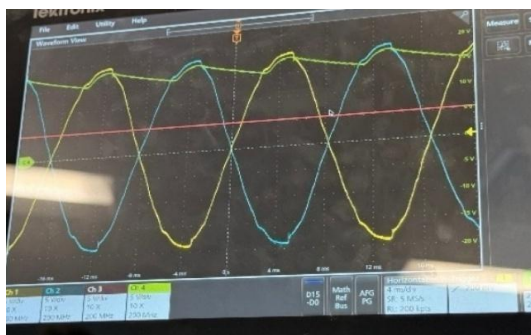


Figure 5.2: Waveform of Figure 5.1 with $100\ \mu\text{F}$ and Zener



Figure 5.3: AC coupling Waveform of Figure 5.1 with $100\ \mu\text{F}$ and Zener

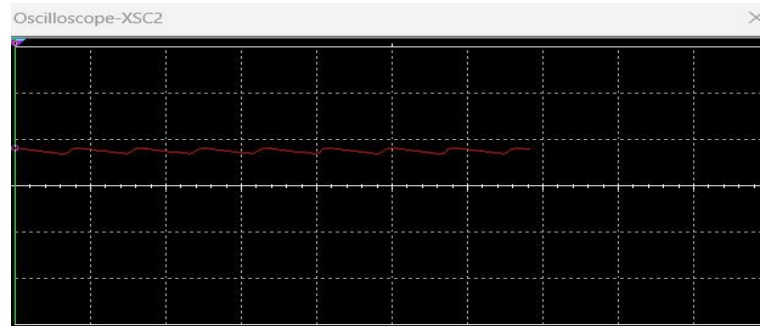
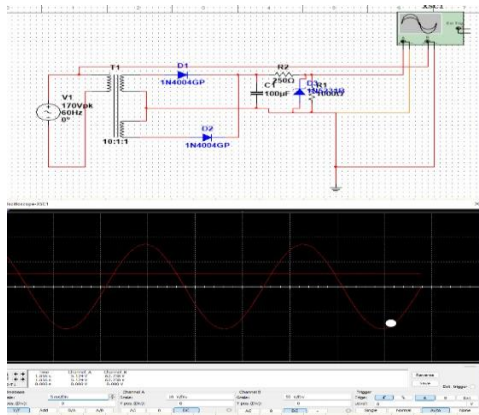


Figure 5.4: Simulation and graph of Figure 5.1 showing VL and VAC1 being on graph

Figure 5.5: Continuation of Figure 5.1 showing graph of V1

W5.2

The experimental waveforms taken during lab are very similar to the Multisim simulation with VAC1 and VAC2 appeared as the expected 60 Hz sine waves with the rectified capacitor voltage V1 showing similar ripple levels as in simulation and in lab. For the load voltage the output appears as a virtually constant $\approx 5.1\text{V}$ both in lab and in the simulation. This indicates that the Zener diode was operating in breakdown and behaving as predicted.

W5.3

$$V_L \approx 5.1\text{V}$$

$$P_{\text{Load}} = \frac{V_L^2}{R_L} = \frac{5.1^2}{1000} = 0.02601\text{W}$$

$$\frac{I_1 - V_L}{250} \cdot V_1 = P_{\text{Source}}$$

$$I_1 = \frac{V_1 - V_L}{R_1} = \frac{17 - 5.1}{250} = 0.0476$$

$$P_{\text{Source}} = V_1 \cdot I_1 = 17 \cdot 0.048 = 0.81$$

$$\eta = \frac{0.0260}{0.81} \cdot 100\% = 3.2\%$$

Figure 5.5 work to calculate experimental efficiency of Figure 5.1

Without the Zener regulator, the output ripple is larger because the capacitor discharges across R_L . With the Zener regulator, the voltage is held at around 5.1V and the measured ripple shown at V_L is much smaller than what was seen in W4-3. We found the Zener regulator was not efficient in this circuit. It had an efficiency of 3.1% due to the amount of excess voltage lost to heat.

$$\eta = \frac{P_{Load}^2}{P_{Source}} * 100, \eta = 3.1\%$$

W6

Circuit Description

The circuit in L6 is a full-wave rectifier built with a center-tapped transformer and two diodes. Each half of the transformer's secondary alternates the polarity. This allows one diode to conduct in the positive half-cycle and the other to conduct during the negative half-cycle, allowing two positive pulses per cycle with a 120 Hz full-wave rectified signal. It is updated from the circuit in L5, as we replaced the Zener diode with an LM7805 linear regulator IC. We also removed the two 510 Ω resistors and replaced them with a 10 Ω "sense resistor" to measure current into the LM7805. This allowed the circuit to supply a steady voltage to power loads.

W6.1

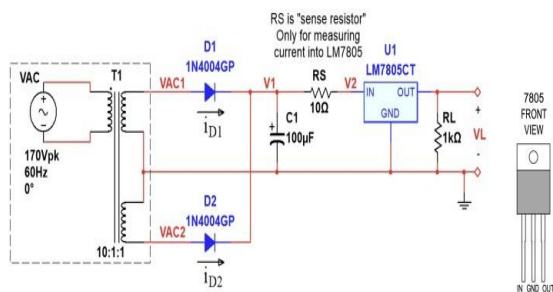


Figure 6.1: Full-Wave Rectifier Circuit
(with cap and LM7805 linear regulator)



Figure 6.2: Waveform of Figure 6.1 with an LM7805

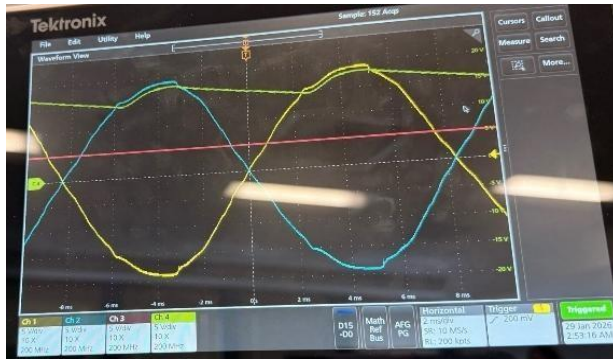


Figure 6.3 Waveform of Figure 6.1 with an LM7805

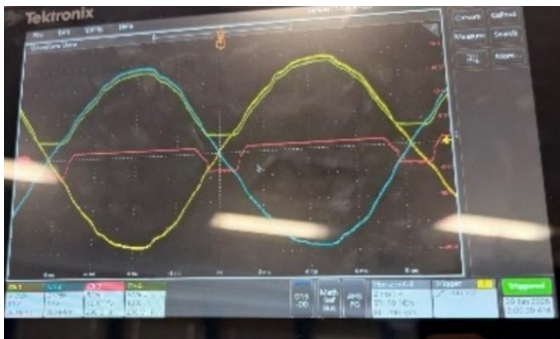


Figure 6.4 Part 1: Waveform of Figure 6.1 with an LM7805



Figure 6.5 Part 2: Waveform of Figure 6.1 with an LM7805

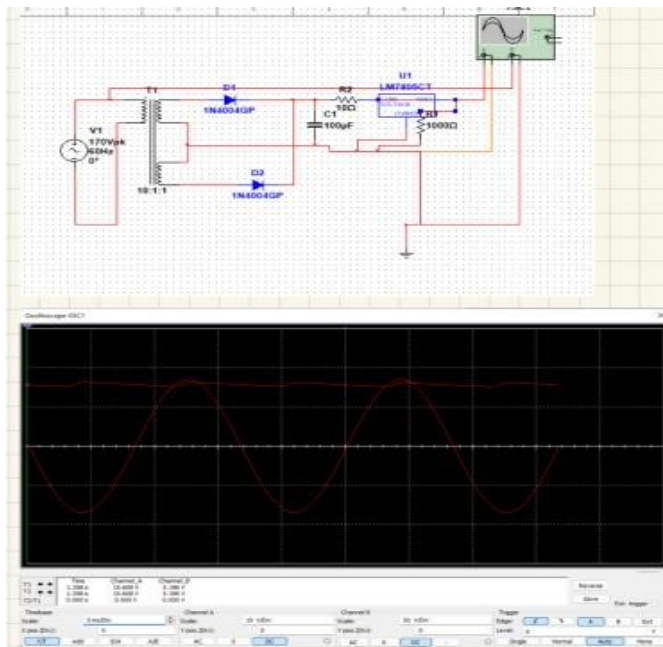


Figure 6.6: Simulation of Figure 6.1

W6.2

The measured waveforms in the LM7805 circuit matched the Multisim simulation well; VAC1 and VAC2 appeared as the expected 60 Hz sine waves. V1 was the full-wave rectified waveform with a small ripple just like the simulation and the LM7805 can regulate it to keep it at about 5V DC. It confirms that the output was properly regulated at 5V with $R_L = 1k\ \Omega$ and $I_L \approx 5\text{ mA}$.

W6.3

The LM7805 proved effective at maintaining a constant 5V output in L6-3, L6-4, and L6-5. There is minimal ripple showing that the regulator has strong load regulation. When the load increased, the input current increased, causing a larger voltage drop across the sense resistor, as expected, but the regulated output remains unchanged. When the load decreases the regulator smoothly reduced its current draw, and keeps the output at 5V which shows it has good stability.

W6.4

Handwritten calculations for efficiency on graph paper:

$$I_S = \frac{(V_1 - V_2)}{R_S} =$$
$$V_1 \approx 18$$
$$V_2 \approx 17.8$$
$$P_{L_{out}} \approx 0.025W$$
$$P_{S_{sense}} = I_S^2 R_S = 0.002$$
$$V_S = 0.02 \cdot 10 = 0.2$$
$$\frac{0.025}{0.36} \approx 6.94\%$$
$$P_{L_{out}} \approx \frac{5^2}{100} = 0.25W \quad V_2 \approx 17.3$$
$$V_{R_S} = I_S \cdot 10 =$$
$$I_S = \frac{V_1 - V_2}{10} = \frac{18 - 17.3}{10} = 0.07A$$
$$P_{I_S} = 1.26W$$
$$\frac{0.25}{1.26} \approx 19.8\%$$

Figure 6.7: Hand Calculations for efficiency

The measured efficiency of the LM7805 regulator was 6.94% for L6-2 with the $1k\ \Omega$ load resistor and 19.8% for L6-4 for the $100\ \Omega$ load resistor. This shows that the regulator was more efficient under a heavier load, as a larger fraction of input power is delivered instead of being dissipated in the regulator and resistor. Compared with the Zener regulator, the LM7805 had higher efficiency. There was an increase in efficiency as the Zener diode continuously dissipated excess current as heat. This reveals that the LM7805 is more efficient and provides better regulation.

W6.5

In this design challenge, we decided to use the 10 μF and the 100 μF capacitors, bringing us two different graphs shown in figure 6.4 and 6.5. To determine the required capacitance, we used the ripple voltage 5V equation for a full-wave rectifier.

The load current was calculated as: $I_{Load} = \frac{5V}{51\Omega} \approx 0.098A$.

Because the LM7805 requires approximately 2V of dropout voltage to remain in regulation, we needed to make the capacitor large enough to prevent the voltage dropping below 7V. The 100 μF capacitor was sufficient as it was the minimum required to fulfill the circuit's needs and reduce the ripple, allowing the LM7805 to maintain a steady 5V output. The 10 μF capacitor failed as it could not regulate the ripple enough. This verified our theoretical prediction.

Discussion

During this lab, DC power supply behavior was analyzed using half-wave rectifiers, full-wave rectifiers, filter capacitors, and voltage regulators. In W1, the half-wave rectifier successfully converted the AC transformer output into a single-polarity waveform. The diode only conducted during the positive half-cycle, causing the load voltage to follow the positive portion of the input while blocking the negative half-cycle. This matched the expected diode behavior, with the output peak was slightly lower due to the diode voltage drop.

In W2, adding a filter capacitor to the half-wave rectifier showed how capacitance affects ripple voltage. With a 10 μF capacitor, the output had a larger ripple because the capacitor discharged more between peaks. As the capacitance increased to 100 μF and then 330 μF , the ripple decreased significantly because the larger capacitor discharged more slowly through the load resistor. The calculated capacitance needed for about 1V peak-to-peak ripple was approximately 333.33 μF , and the 330 μF capacitor closely matched this prediction.

In W3 and W4, the full-wave rectifier improved the output by producing two positive pulses per AC cycle. Since the capacitor was recharged twice as often, the ripple frequency doubled to 120 Hz, making the output smoother than the half-wave rectifier for the same load and capacitance. The 200 μF capacitor in the full-wave circuit reduced the ripple to around the desired 1V peak-to-peak range, which matched the theoretical prediction that a smaller capacitor is needed for full-wave filtering compared to half-wave filtering.

In W5, the Zener diode regulator held the load voltage at approximately 5.1V and reduced the ripple compared to the unregulated full-wave rectifier. This showed that the Zener diode was operating in breakdown and clamping the output voltage as expected. However, the Zener regulator had very low efficiency, around 3.1%, because excess voltage and current were dissipated as heat in the series resistor and Zener diode.

In W6, the LM7805 linear regulator provided a more stable 5V output and showed better load regulation than the Zener circuit. The output remained close to 5V even as the load changed, confirming that the regulator was operating properly. The measured efficiency improved compared to the Zener regulator, reaching about 6.94% with the 1k Ω load and 19.8% with the heavier 100 Ω load. The design challenge also showed the importance of the input capacitor size, since the 10 μ F capacitor could not keep the regulator input above the dropout voltage, while the 100 μ F capacitor was sufficient to maintain regulation. Overall, the results closely matched the theoretical predictions and simulations, with minor differences likely due to diode drops, capacitor tolerances, regulator dropout behavior, and measurement limitations.

Conclusion

In conclusion, this lab demonstrated how AC voltage can be converted into a more stable DC output through rectification, filtering, and regulation. The half-wave and full-wave rectifiers showed how diodes control which parts of the AC waveform reach the load, with the full-wave rectifier producing a smoother output because it recharges the filter capacitor twice per cycle. We found that increasing the filter capacitance reduced ripple voltage, closely matching the expected relationship between ripple, capacitance, load resistance, and frequency. The Zener regulator was able to hold the output near 5.1V and reduce ripple, but it had low efficiency because excess power was dissipated as heat. The LM7805 regulator produced a more stable 5V output and handled load changes better, with improved efficiency compared to the Zener circuit. Some small differences between measured and theoretical values were likely due to diode voltage drops, capacitor tolerances, regulator dropout behavior, and measurement limitations. This lab reinforced concepts such as rectification, ripple reduction, voltage regulation, load regulation, and the design of stable DC power supplies.